Consider three bases for $\mathbb{R}^2$

$$\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}\right\}, \left\{\begin{pmatrix}2\\3\end{pmatrix},\begin{pmatrix}4\\5\end{pmatrix}\right\}, \left\{\begin{pmatrix}1\\5\end{pmatrix},\begin{pmatrix}-5\\1\end{pmatrix}\right\}$$

First one is easiest to work with, standard basis after all. But if forced would we rather work with second one or third one? Doesn't immediately seem like much difference.

But there actually is! In third the two basis elements are orthogonal to each other! That actually makes big difference (second basis doesn't have that). And it's also related to why standard basis is easier, as in any $\mathbb{R}^n$ every standard basis element is orthogonal to every other.

So that issue is quite relevant - relates to next content.

Last time talked about subspace W of $\mathbb{R}^m$ has orthogonal complement $W^\perp$ consisting of all vectors which are orthogonal to W (that is, orthogonal to every vector in W).

How do we "find" $W^\perp$? When we say find, the natural thing is to try find a basis for it as we would with any subspace. Well since we "know" what W is, presumably we have a basis for W so we start there.

Say W is a subspace of dimension n in $\mathbb{R}^m$ (so $n \leq m$)

Put the basis for W as the columns of an $m \times n$ matrix (or the rows of an $n \times m$ matrix, just the transpose, no fundamental difference).

Thinking of it as the latter, the set of vectors orthogonal to all those basis rows is the null space of the matrix.

I.e. recall last time we mentioned that for any real matrix A that we have

$$(\text{Row } A)^{\perp} = \text{Nul } A$$

So we find the null space of that mentioned matrix and that gives us the desired basis for $W^{\perp}$

Hence that's a way of viewing/finding $W^{\perp}$.

6.2 Orthogonal Sets

In a vector space V a set of vectors is an <u>orthogonal set</u> if each vector is orthogonal to every other vector in the set.

Note that we often assume (not required) that such a set has only nonzero vectors. While the 0 vector fits in well in an orthogonal set since it is orthogonal to every vector, the 0 vector also works quite differently so we generally don't want to allow it for various reasons.

Also note that we have only defined orthogonal via dot product thus far so mostly assuming V is a subspace of some $\mathbb{R}^m$. But via inner product can perhaps define this for other vector spaces using same idea. That comes later though.

If such an orthogonal set is a basis for V, we call it an <u>orthogonal basis</u>. Hence one reason we generally don't

want the 0 vector in an orthogonal set as a set with 0 would never be a basis for a nontrivial vector space.

One important thing is that an orthogonal set of nonzero vectors is always linearly independent (yet another reason we prefer the 0 vector not to be in an orthogonal set as then this would not be true).

Why is this so? Say set of nonzero $\{u_1, u_2, \ldots, u_n\}$ is an orthogonal set but is linearly dependent. Then some vector in it, say without loss of generality u_n, is a linear combination of others, i.e. we have

$$u_n = c_1 u_1 + c_2 u_2 + \ldots + c_{n-1} u_{n-1}$$

Dotting each side with u_n and then noting orthogonality yields

$$u_n \cdot u_n = 0$$

---->

$$u_n = 0$$

And this is a contradiction since this orthogonal set was stated not to have the 0 vector.

One way orthogonal helps: Say W is a subspace of $\mathbb{R}^m$ with basis $\{u_1, u_2, \ldots, u_n\}$. Then for any $y \in \mathbb{R}^m$ we, by definition of basis, can always write

$$y = c_1 u_1 + c_2 u_2 + \ldots + c_n u_n$$

for unique scalars $c_1, c_2, \ldots, c_n$. But how to find these scalars? In theory have to solve linear system

$$\underbrace{\begin{bmatrix} u_1 & u_2 & \ldots & u_n \end{bmatrix}}_{\text{columns}} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = y$$

This is an n x n system. Unless n is small, this could be quite long! But we have a nice shortcut if the stated basis is an orthogonal basis. That is, if that is true

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j} \qquad \text{for all} \quad 1 \le j \le n$$

Ok still have to do this n times, but still that's far better than solving it via row reduction or inverse matrix. Why does this work? For any such j

$$y \cdot u_j = (c_1 u_1 + c_2 u_2 + \ldots + c_n u_n) \cdot u_j$$

Since $u_i \cdot u_j = 0$ for all $i \neq j$ (orthogonal basis), this becomes (distributive law)

$$y \cdot u_j = c_j (u_j \cdot u_j)$$

which solves to what we mentioned.

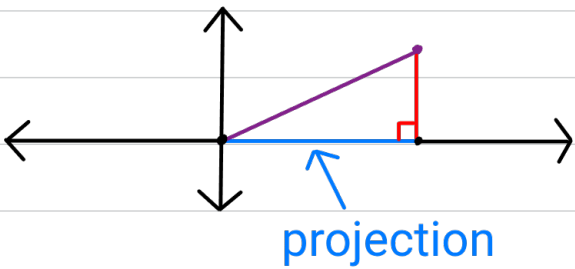
We speculated earlier as to which of these bases for $\mathbb{R}^2$ we would rather work with:

$$\left\{\begin{pmatrix}2\\3\end{pmatrix},\begin{pmatrix}4\\5\end{pmatrix}\right\} \quad , \quad \left\{\begin{pmatrix}1\\5\end{pmatrix},\begin{pmatrix}-5\\1\end{pmatrix}\right\}$$

Neither is standard basis so might seem fairly equally difficult to deal with. And indeed in some contexts that probably would be true, e.g. if you wanted to compute say a change of basis matrix involving either, likely would be similar.

But in other contexts, like solving the type of system just mentioned, the fact that the second basis is is orthogonal will make it much easier to deal with (will see other contexts where this is the case later in this chapter as well).

Also use orthogonality for <u>orthogonal projections</u> (or projections for short).



Projecting vector in purple on subspace, the x-axis.

What does this mean? Many ways to explain. Perhaps easiest to understand in low dimensions is the point in subspace which is closest to the outside point. Found via drawing perpendicular from point to subspace.

But how do we find this projection?

In $\mathbb{R}^2$ projection of point on some line, use slope to help find that, fairly straightforward. In $\mathbb{R}^3$ projection of point on plane usually, that's definitely more work to find, maybe use cross product.

Say we were in $\mathbb{R}^{10}$ and took a seven dimensional subspace. For some point not in the subspace could you still find a point in subspace closest to that outside point?

Well main key is idea of length/distance formula works in any dimension so in that regard it seems likely. But since three dimensional is already quite a bit tougher than two, how would we do that?

That's main point of this, understanding how we would attack such a problem. Sure such problems often won't be practical to actually do by hand - understanding the ideas is important for one so you can do it in situations where it's reasonable (in say $\mathbb{R}^4$ sometimes it could could be and your prior known ideas won't work there).

But also if we wanted to teach a machine to do this how to ensure they do these calculations as efficiently as possible could be a relevant thing as well.

To start: in any $\mathbb{R}^m$ take a nonzero vector u and an arbitrary vector y. We decompose y into the sum of a multiple of u and a vector orthogonal to u.

The first of these is the projection of y on u, denoted $\hat{y}$ or $\text{proj}_L y$ where L is the subspace generated by u.

We use the former notation more, though the latter, while more clunky, is perhaps more clear since shows what we are projecting y on, $\hat{y}$ notation doesn't do that, context really has to make that very clear.

This type of projection is what we were talking about earlier with closest point, just restricted setting as we are projecting y on a subspace generated by a single vector, i.e. a one dimensional subspace.

Next section we will generalize to projecting on higher dimensional subspaces. Useful to do it for one one dimensional first since helps get a feel for it, easier naturally for lower dimensional, and generalizes with a bit more work into higher dimensional.

How to compute this projection? Formula is

$$\hat{y} = \frac{y \cdot u}{u \cdot u} u$$

Be careful with this, don't mix up dot product of two vectors with scalar multiplying a vector.

Why does this work? Well clearly $\hat{y}$ is a multiple of u, so that's no issue. Other thing we need is for $y - \hat{y}$ to be orthogonal to u. To that end, see if its dot product with u is guaranteed to be 0. Checking:

$$(y - \hat{y}) \cdot u = y \cdot u - \left[\frac{y \cdot u}{u \cdot u} u \right] \cdot u$$

$$= y \cdot u - y \cdot u$$

$$= 0$$

Proving $y - \hat{y}$ is always orthogonal to u, as desired.

An <u>orthonormal</u> set/basis is an orthogonal set/basis with the additional condition that every vector has length 1.

Can be a bit easier, e.g. in above with $u \cdot u$, if that is always 1, saves time from computing it every moment.

Fortunately it's quite easy to switch from orthogonal to orthonormal, just replace each vector u with $\frac{u}{\| u \|}$

Doing that doesn't change direction of any vector so stays orthogonal, just now every vector has length 1.

Since switching is easy we often just go to orthogonal, knowing that switching to orthonormal is no issue.

This is yet another reason why standard basis is generally easier as in any $\mathbb{R}^m$ it is always an orthonormal basis.

Orthonornal helps in many ways. A good start is a real matrix U (not necessarily square matrix) has its columns being an orthonormal set exactly when

$$U^T U = I$$

This is quite clear since every diagonal entry in $U^T U$ is the dot product of some column of U with itself, which is 1 due to orthonormal.

And every off diagonal entry is the dot product of some column of U with a differen column, which is 0 since since orthonormal includes orthogonal. Hence that product is the identity matrix.

One interesting property this has is for any vector x of proper size we have:

$$\| Ux \| = \| x \|$$

The fact that this holds even though Ux and x are not in same $\mathbb{R}^m$ if U is not square is quite unexpected.

How to prove?

$$\| Ux \| = \| (Ux)^T \| = \| x^T U^T \|$$

With this in mind, note further

$$\| x^T U^T U x \| = \| x^T I x \| = \| x^T x \| = \| x \|^2$$

So breaking the first term of this up means

$$\| x^T U^T \| = \| Ux \| = \| x \|$$

as the first two just above we proved equal. Latter two is what we wanted to prove.

Could also do this brute force. If $U = [u_{ij}]$ and $x = [x_j]$, the coefficient of say x_1^2 in $\| Ux \|$ will be

$$u_{11}^2 + u_{21}^2 + \ldots + u_{m1}^2$$

which becomes 1 since U has orthonormal columns, same as it is in $\| x \|$, which is what is needed to prove $\| Ux \| = \| x \|$

So say you have an orthonormal set/basis in a space. If you multiply each vector by such a U, since that preserves length, vectors will stay of length 1 in new space which could be valuable.

Since orthornormal definitely makes things easier than orthogonal in many settings, maybe should just use orthonormal only all the time?

Well not necessarily, say we have some generic vector like (7, 8, 9) which is part of some orthogonal set. If we scale that to make it have length 1, that will involve dividing by an annoying square root.

So the calculations that might be involved then especially since the other vectors in the orthogonal set might be similar looking could actually be much worse!

Thus orthonormal may not always be better than orthogonal, definitely depends on the situation, you have to use your judgement on that.

Yet another reason why standard basis is nice, it is an orthonormal basis, and also has integer entries, not coming with annoying square roots that would arise if you try to normalize an orthogonal set which had a vector of the type (7, 8, 9).

Special case of matrix with orthonormal columns is a square matrix which has that property. This is called an <u>orthogonal matrix</u>.

A square matrix U is orthogonal exactly when $U^T = U^{-1}$

Yet another place you see transpose come up.

Remember this is called orthogonal matrix even though columns are orthonormal, not just orthogonal. Just is the convention, again perhaps because switching from an orthogonal set of vectors to an orthonormal set is easy.

Orthogonal matrices are naturally quite convenient for calculation based reasons as the fact that their columns (and rows for that matter) are pairwise orthogonal, each column having length 1, makes alot of the calculations you might otherwise have to do with generic matrices much more convenient.

As such, you will be seeing more of these later.

6.2.37 If U and V are orthogonal matrices of same size then UV is orthogonal as well. Why?

$$(UV)^T = V^T U^T = V^{-1} U^{-1} = (UV)^{-1}$$

That proves it. Fact that orthogonal matrices behave well under product is one reason they are useful.

Similar to why determinants were so helpful since they also behaved well under product.

6.3 Orthogonal Projections

Want to now discus concept of projecting vector onto subspace of perhaps dimension higher than one.

For W some nontrivial subspace of $\mathbb{R}^m$ and $y \in \mathbb{R}^m$ we want to write y in form:

$$y = \hat{y} + z$$

where $\hat{y} \in W$ (this is the projection of y on W) and $z \in W^{\perp}$. Two immediate questions here

(1) How do we know such a breakdown always exists?
(2) If such a breakdown exists, is it always unique?

To help us with (1) take $\{u_1, u_2, \dots, u_n\}$ an orthogonal basis for W.

Similarly as earlier we have

$$\hat{y} = \sum_{i=1}^{n} \frac{y \cdot u_i}{u_i \cdot u_i} u_i$$

Just a sum now of terms of that type rather than a single one. This highlights much better why we might want orthonormal basis as every $u_i \cdot u_i$ being 1 can save us a lot of calculation, not just one brief thing as earlier when there was only a single term.

Very important - do not apply this formula unless the basis is orthogonal!! That is a very common mistake I've seen in the past.

I.e., when asked to find the projection of a point on a subspace, folks just use an arbitrary basis. E.g. if projecting on the column space of some matrix, they use the pivot columns, which are a basis for Col A, but may well not be an orthogonal basis.

Or if given a basis they just use it directly without converting it to orthogonal (converting is what next section is about). That will be a big problem as that will almost certainly give the wrong answer!

So why does this formula work? Similar to earlier again.

$\hat{y} \in W$ is obvious as it is a linear combination of basis elements of W. Really only need to show $y - \hat{y} \in W^{\perp}$

To that end, note that for any j in $[1, n]$:

$$(y - \hat{y}) \cdot u_j = y \cdot u_j - \left(\sum_{i=1}^{n} \frac{y \cdot u_i}{u_i \cdot u_i} u_i \right) \cdot u_j$$

Just as earlier, $u_i \cdot u_j = 0$ whenever $i \neq j$

So this becomes

$$(y - \hat{y}) \cdot u_j = y \cdot u_j - \left[\frac{y \cdot u_j}{u_j \cdot u_j} u_j \right] \cdot u_j$$

$$= y \cdot u_j - y \cdot u_j$$

$$= 0$$

Thus $y - \hat{y}$ is orthogonal to every u_j. Hence

since the u_j's are a basis for W, $y - \hat{y}$ is

orthogonal to every vector in W, proving $y - \hat{y} \in W^{\perp}$

So this proves such a breakdown is always possible.

This helps with many things, like finding point in subspace closest to some outside point.

Will discus that more later as well as proving the uniqueness of this breakdown next time.